THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 6	Common prime factors give HCF $= 6$
2	**(d)** 6	Substitute $x = 4$: $p(4) = 6$
3	**(a)** infinitely many solutions (coincident lines)	$\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{1}{2}$, $\frac{c_1}{c_2} = \frac{1}{2}$
4	**(a)** $x^2 + 3x + 2 = 0$	Degree must be 2
5	**(c)** 12	LCM $= \dfrac{4 \times 6}{2} = 12$
6	**(c)** 4	$2x - 8 = 0 \Rightarrow x = 4$
7	**(a)** Yes	$2(3) + 3(2) = 12$
8	**(a)** Yes	LHS $= 0$ at $x = 2$
9	**(b)** 3	$12 = 2^2 \times 3$
10	**(b)** 5	Sum of zeroes $= -b/a$
11	**(c)** 7	$k = 2(2) + 3(1)$
12	**(b)** -3	Compare with $ax^2 + bx + c = 0$
13	**(b)** $\sqrt{7}$	7 is not a perfect square, so $\sqrt{7}$ is irrational
14	**(b)** 6	Product of zeroes $= c/a$
15	**(a)** 7	$x + y = 10$, $x - y = 4$ gives $2x = 14$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$18 = 2 \times 3^2$, $24 = 2^3 \times 3$ (1); HCF $= 6$ (1).	1+1
17	$p(3) = 10$ (2).	2
18	$\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{1}{2}$ (1); all ratios equal, infinitely many solutions, consistent (1).	1+1
19	Put $x = 2$: LHS $= 0$ (1); equals RHS 0, so it is a root (1).	1+1
20	Divide repeatedly by primes (1); $140 = 2^2 \times 5 \times 7$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$26 = 2 \times 13$, $91 = 7 \times 13$ (1); HCF $= 13$ (1); LCM $= 182$ (1).	1+1+1
22	Factors $(x - 3)(x - 5)$ (1); zeroes $3, 5$ (1); sum $8 = -b/a$ verified (1).	1+1+1
23	Multiply (i) by -1 and (ii) by 1 and subtract: $(-2)x = -10$ (2); $x = 5$ (1); put in (i): $y = 3$ (1); check in (ii) (1).	2+1+1+1
24	Split the middle term: $-5 = -3 - 2$ because $ac = 6$ (1); regroup and factorise (1); roots $x = 2, 3$ (1).	1+1+1
	OR: HCF $= 6$, LCM $= 60$ (1+1); HCF×LCM $= 360 = 12 \times 30$ (1).	1+1+1
25	$p(x) = k[x^2 - (\text{sum})x + \text{product}]$ (1); $x^2 - 5x + 6$ (1); one such polynomial is $x^2 - 5x + 6$ (1).	1+1+1
	OR: From (i): $y = \dfrac{18 - 2x}{3}$ (1); substitute in (ii) (1); simplify to $(1)x = 3$ (1); $x = 3$ (1); $y = 4$ (1).	1+1+1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$135 = 3^3 \times 5$, $225 = 3^2 \times 5^2$ (2); HCF $= 45$ (1); LCM $= 675$ (1); $45 times 675 = 30375$ verified (1).	2+1+1+1
	OR: Zeroes 2, 3 (2); sum $= 5 = -\dfrac{b}{a}$ (1.5); product $= 6 = \dfrac{c}{a}$ (1.5).	2+1.5+1.5
27	Multiply (i) by 4 and (ii) by 3 and subtract: $(-7)x = -21$ (2); $x = 3$ (1); put in (i): $y = 2$ (1); check in (ii) (1).	2+1+1+1
	OR: Split the middle term: $-7 = -6 - 1$ because $ac = 6$ (1); regroup and factorise (1); roots $x = \frac{1}{2}, 3$ (1).	1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 2 and 3	Put each factor equal to zero
2	**(b)** $2^2 \times 3 \times 5$	Divide repeatedly by primes: $60 = 2^2 \times 3 \times 5$
3	**(d)** 2	A polynomial of degree n has at most n zeroes
4	**(b)** y-axis	All points have the same x-coordinate
5	**(d)** 1	$D = b^2 - 4ac$
6	**(a)** terminating	Denominator 8: only 2 and 5 as prime factors
7	**(a)** Yes	$p(2) = 0$
8	**(a)** coincident	One equation is a multiple of the other
9	**(a)** two distinct real roots	$D = b^2 - 4ac = 1$
10	**(b)** 10	Common prime factors give HCF $= 10$
11	**(b)** 16	Substitute $x = 3$: $p(3) = 16$
12	**(a)** no solution (parallel lines)	$\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{1}{2}$, $\frac{c_1}{c_2} = \frac{5}{7}$
13	**(c)** $x^2 - 5x + 6 = 0$	Degree must be 2
14	**(c)** 18	LCM $= \frac{6 \times 9}{3} = 18$
15	**(a)** -3	$3x + 9 = 0 \Rightarrow x = -3$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$3x - 12 = 0$ (1) so $x = 4$ (1).	1+1
17	Adding: $2x = 16$ (1) so $x = 8$, $y = 4$ (1).	1+1
18	$x - 4 = 0$ or $x - 5 = 0$ (1); $x = 4$ or $x = 5$ (1).	1+1
19	$16 = 2^4$ (1); it has only 2 and/or 5, so terminating (1).	1+1
20	Factorise: $(x - 2)(x - 5)$ (1); zeroes 2 and 5 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$a = 2, b = -4, c = 3$ (1); $D = -8$ (1); no real roots (1).	1+1+1
22	$16 = 2^4$ (1); terminating (1); value ≈ 0.1875 (1).	1+1+1
23	Put $x = 1$: $p(1) = 0$ (2) $= 0$, hence a zero (1).	2+1
24	Let numbers be x, y: $x + y = 25$, $x - y = 11$ (1); adding $2x = 36$ (1); numbers 18 and 7 (1).	1+1+1
	OR: Let integers be x and $x + 1$: $x(x + 1) = 20$ (1); $x^2 + x - 20 = 0$ (1); $(x - 4)(x + 5) = 0$, $x = 4$ (1); integers 4 and 5 (1).	1+1+1+1
25	$96 = 2^5 \times 3$, $404 = 2^2 \times 101$ (1); HCF $= 4$ (1); LCM $= 9696$ (1).	1+1+1
	OR: Factors $(x + 2)(x - 6)$ (1); zeroes $-2, 6$ (1); sum $4 = -b/a$ verified (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$12 = 2^2 \times 3$, $18 = 2 \times 3^2$, $24 = 2^3 \times 3$ (3); HCF $= 6$ (1); LCM $= 72$ (1).	3+1+1
	OR: Sum $= 6 = -\frac{b}{a}$ (2); sum of products in pairs $= 11 = \frac{c}{a}$ (2); product $= 6 = -\frac{d}{a}$ (1).	2+2+1
27	Let pencil $= x$, eraser $= y$: $2x + 3y = 22$, $3x + 2y = 23$ (2); solving, $x = 5$, $y = 4$ (2); pencil ₹5, eraser ₹4 (1).	2+2+1
	OR: Breadth x, length $x + 5$: $x(x + 5) = 66$ (1); $x^2 + 5x - 66 = 0$ (1); $(x - 6)(x + 11) = 0$ (1); $x = 6$ m (breadth), length 11 m (reject negative) (2).	1+1+1+2

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** Yes	$3(2)+1(5)=11$
2	**(d)** No	LHS $=2$ at $x=4$
3	**(c)** 3	$30=2\times3\times5$
4	**(d)** 4	Sum of zeroes $=-b/a$
5	**(c)** 14	$k=3(4)+1(2)$
6	**(a)** 7	Compare with $ax^2+bx+c=0$
7	**(d)** $\sqrt{11}$	11 is not a perfect square, so $\sqrt{11}$ is irrational
8	**(a)** -5	Product of zeroes $=c/a$
9	**(b)** 10	$x+y=14$, $x-y=6$ gives $2x=20$
10	**(b)** -1 and 4	Put each factor equal to zero
11	**(a)** $2^2\times3\times7$	Divide repeatedly by primes: $84=2^2\times3\times7$
12	**(c)** 3	A polynomial of degree n has at most n zeroes
13	**(a)** y-axis	All points have the same x-coordinate
14	**(b)** 1	$D=b^2-4ac$
15	**(a)** non-terminating repeating	Denominator 15: has a prime factor other than 2 and 5

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$k=3(2)+2(5)\,(1)=16$ (1).	1+1
17	$D=(-6)^2-4(1)(8)=4$ (1); two distinct real roots (1).	1+1
18	$30=2\times3\times5$, $45=3^2\times5$ (1); HCF $=15$ (1).	1+1
19	$p(-2)=0$ (2).	2
20	$\frac{a_1}{a_2}=\frac{1}{2}$, $\frac{b_1}{b_2}=\frac{1}{2}$ (1); $a_1/a_2=b_1/b_2\neq c_1/c_2$, no solution, inconsistent (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Multiply (i) by -1 and (ii) by 1 and subtract: $(-3)x=-12$ (2); $x=4$ (1); put in (i): $y=2$ (1); check in (ii) (1).	2+1+1+1
22	Split the middle term: $-4=-5+1$ because $ac=-5$ (1); regroup and factorise (1); roots $x=-1,5$ (1).	1+1+1
23	HCF $=7$, LCM $=70$ (1+1); HCF×LCM $=490=14\times35$ (1).	1+1+1
24	$p(x)=k[x^2-(\text{sum})x+\text{product}]$ (1); x^2+3x+2 (1); one such polynomial is x^2+3x+2 (1).	1+1+1
	OR: From (i): $y=\dfrac{12-1x}{3}$ (1); substitute in (ii) (1); simplify to $(5)x=30$ (1); $x=6$ (1); $y=2$ (1).	1+1+1+1+1
25	$a=1$, $b=-8$, $c=16$ (1); $D=0$ (1); two equal real roots (1).	1+1+1
	OR: $25=5^2$ (1); terminating (1); value ≈0.28 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$196=2^2\times7^2$, $280=2^3\times5\times7$ (2); HCF $=28$ (1); LCM $=1960$ (1); $28 times 1960=54880$ verified (1).	2+1+1+1
	OR: Zeroes -4, 1 (2); sum $=-3=-\dfrac{b}{a}$ (1.5); product $=-4=\dfrac{c}{a}$ (1.5).	2+1.5+1.5
27	Multiply (i) by -3 and (ii) by 2 and subtract: $(-13)x=-52$ (2); $x=4$ (1); put in (i): $y=1$ (1); check in (ii) (1).	2+1+1+1
	OR: Split the middle term: $-1=-3+2$ because $ac=-6$ (1); regroup and factorise (1); roots $x=-\frac{2}{3},1$ (1).	1+1+1

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